

Healthy Talks: Facilitating Collective Speculations and Co-design for Local Content Services

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ABSTRACT

Despite 800 million additional individuals joining the Internet between 2019-2021, a significant digital divide persists. Moreover, monopoly over Internet Infrastructure has led to people having little control over their data. Community Networks (CNs) are an emerging solution providing autonomy and safe spaces in the digital context. While CNs can improve accessibility to the Internet, another critical component is local content, which makes the Internet more relevant for users. This paper explores methods and practices that can be adopted to promote local content production and dissemination. The research/study was conducted with the local front-line health workers (Health Navigators/HNs), the primary content creators of the COWHKI initiative. The Co-design and Speculative design activities reveal insights about the challenges and experiences of the HNs in engaging with content and technology. These insights have been translated into recommendations for Papad, the digital interface used for creating and curating content in the COWHKI initiative.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; User studies; Human computer interaction (HCI); HCI design and evaluation methods; Field studies; Interaction design.

KEYWORDS

Community Networks, Co-design, Democratisation of the Internet, Digital Divide, Local Content, Speculative design

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1 INTRODUCTION

The Internet is one of the most used primary avenues for information transmission and communication in the modern world. However, despite 800 million additional individuals joining the Internet between 2019 and 2021 [7], only 43% of Indians used the Internet in 2020 [17]. To direct advancements in Internet connectivity and

reduce the social ramifications of a digital divide, the Indian government released the Broadband Policy for the Government of India (2004) and the National Telecommunications Policy (2012). It has also proposed and undertaken several projects and schemes which have met with varying levels of success. The Supreme Court of India further enforces this by declaring Internet access as a fundamental right by extension [12]. Nonetheless, notwithstanding this judgement and other advances, a significant digital divide restricts segments of society from gaining access to the Internet. On the other side of the Internet connectivity problem, the past two decades of growth and development in ICTs have seen strategic privatisation and monopoly over Internet infrastructure, with people having little control over how data and information are stored and used [6].

Community Networks (CNs) offer a potential solution, as they are community-led and owned, allowing for autonomy and safe spaces in the digital context [15]. One such group is the Community Owned Wireless Health Knowledge Infrastructure (COWHKI) in Channapatna, Karnataka, India, which is currently testing various approaches to network ownership, management, and content services. The main work of COWHKI is an archive of local health knowledge and practices called the Channapatna Health Library (CHL), accessed using an app called Papad [10].

This study examines ways to promote the production and distribution of locally generated content by the community through the COWHKI initiative. A co-design and a speculative design exercise were conducted to develop content and envision potential developments for Papad and the CHL. The research reveals insights into how stakeholders interact with content and technology.

2 RELATED WORK

2.1 Local content in CNs

While CNs can effectively improve accessibility to the Internet, another critical component in bridging digital divides is local content, which makes the Internet more relevant and attractive for users [8, 16]. Local content is locally important, locally generated, or both. Echaniz and Peze from AlterMundi, an NGO based in Argentina working on wireless CNs for rural areas, throw light on the limitations of just 'providing access' to reduce digital divides. They claim this inadvertently supports power concentration in internet ecosystems by fostering a unidirectional relationship with the end users [5]. In a manifestation of the right to co-create the Internet, people should be viewed as empowered citizens who utilise the ability to produce, host and control their content to address local communication challenges and share their local culture effectively. Local alternative services hosted on CNs often find it hard to compete directly with global corporations and systems. CNs thus have

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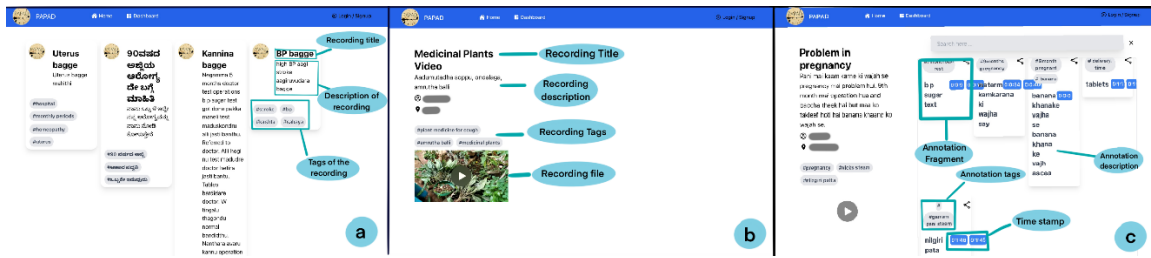


Figure 1: Screenshots from Papad interface (a) recordings within the CHL (b) recording without annotations (c) recording with annotated fragments

a responsibility to strengthen local services beyond just internet access.

2.2 Community-led initiatives with a focus on content creation

With digital transformations becoming commonplace, several initiatives took this as an opportunity to set up CNs or similar models across India (and worldwide) and empowered communities and local self-governments to take responsibility for maintaining and prioritising digital services important locally.

One of the first ever CNs to be set up in India was in Chanderi, Madhya Pradesh [9]. Weaving is the primary livelihood for the people of Chanderi. Some of the solutions envisioned for the problems faced by weavers were creating a digital resource centre where the weavers could store, tag, and archive traditional and new weaving designs and patterns and launching an e-commerce site. The CN in Chanderi has helped revive the handloom industry and increased the income of the weavers [9].

GramVaani is a non-profit endeavour based in India working on using technology to create participatory media platforms for rural communities [14]. Using these IVR platforms, GramVaani has provided rural India with a way to engage and create content regarding women's health, sexual and reproductive health, workplace harassment and more. Their technology allows users to leave a missed call and then get a call back (free of cost) with which they can record their own content or listen to pre-recorded ones [14].

3 COMMUNITY OWNED WIRELESS HEALTH KNOWLEDGE INFRASTRUCTURE (COWHKI)

About 80 kilometres from Bangalore, India, is the town of Channapatna, where Collectives Design Beku [4], Janastu [11], and MAYA Health [13] collaborate to improve community well-being. The result of this collaboration is the initiative called Community Owned Wireless Health Knowledge Infrastructure (COWHKI). The COWHKI initiative is working on co-developing and maintaining a digital safe space for the community to store, tag and archive their own data and knowledge related to health and well-being. This archive is called the Channapatna Health Library (CHL). The CN at Channapatna gives access to a server that hosts and runs Papad, the primary tool for collecting and annotating audio and video artefacts about local health practices. It aims to help communities store and share content about topics that are important to them. Content

in Papad is in audio and video formats to be more accessible for populations with basic literacy levels. It is organised using tags and is annotated.

At the heart of the COWHKI project lies the front-line health workers from MAYA Health (called Health Navigators or HNs), who are a group of driven and aspirational women from marginalised sections of the community. The HNs perform screening tests, check in with the clients about their prescriptions and provide dietary and lifestyle advice. They also collect audio-video artefacts about local health knowledge and practices and upload them to Papad, where they tag and add annotations. Through the COWHKI project, the HNs have been building and curating the CHL and aim to move into the role of local content creators as they co-own, maintain and manage the CN infrastructure and the CHL.

4 FIELD ENGAGEMENTS

The topic of interest for this study was Local Content Services. As stated earlier (in section 2.1), to fully realise the right to co-create the Internet, the key is to produce and host local content and service to solve local communication challenges efficiently. Locally relevant and locally created content serves the community's needs and provides a safe space to interact. The study's main aim was to understand the challenges the HNs face in accessing, synthesising and creating relevant content. The research was conducted through weekly discussions and activities between September and November 2022.

4.1 Methods

The research was conducted via conversations, 'fly-on-the-wall' observations, and small activities based on speculative design and co-design. These were carried out with the field coordinator, the HNs and their clients (suitable methods were chosen for each group). The HNs are the primary actors in aiding the community (mainly the elderly, women and children) to engage with content meaningfully. To better understand what the HNs envisioned for their role as content creators, further research was conducted through two focused activities with interested HNs.

4.1.1 Co-designing supporting material for the CHL.. A repeated sentiment amongst the HNs was the need for physical posters, text, and visuals to support the audio library and distribute it to their clients. One motivation behind this was that the CN infrastructure was currently not accessible from areas other than the MAYA Health main building, and a significant portion of the community did not

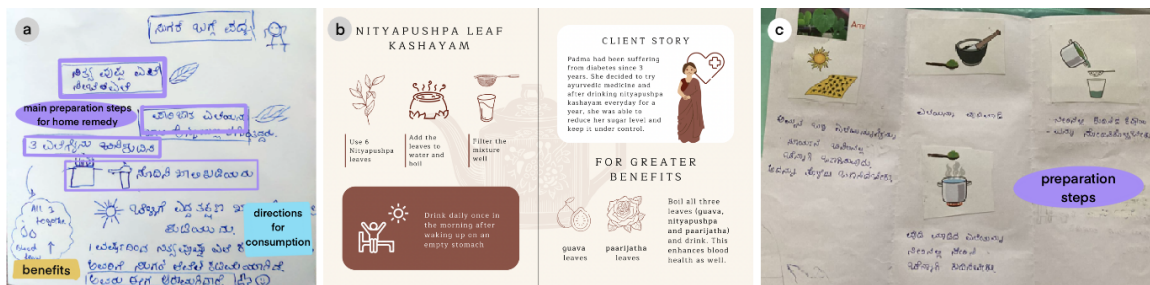


Figure 2: (a) visual notes by HN1 in Kannada (b) English version of pamphlet made based on visual notes (c) HN1's pamphlet in Kannada

have the means to access digital content. Thus, the HNs and the author felt that simple flyers and posters about different audios in the CHL would help generate interest in both the advice the HNs give and the CHL. During a one-on-one session with HN1, visual notes on an audio recording she had collected were made (Figure 2 (a)), based on which a simple pamphlet was created (Figure 2 (b)). HN1 shared the pamphlet with her clients seeing which other HNs also showed interest. Over the subsequent two sessions, elements that would help the creation of such a pamphlet were discussed and designed, leading to the creation of a toolkit that they could use to produce content by themselves. These materials in the toolkit are described below.

• Icons relating to food, body parts, timings etc.

The first thing that came up during the discussions was using images.

HN1: "I feel that my drawing skills are not good enough. Also, even if I draw, I do not think it will look as nice as the ones made and printed using a computer. But I still want to try more today. I have another recording, and maybe together, another poster can be made for the client. . . I could try on my own also if I did not have to draw." Some other HNs agreed with her that having a simple set of images to start with would make more HNs willing to try.

• Layout and Style Elements

While observing HN1 make her visual notes, it was observed that she found it easier to start when the author drew in a few placeholder boxes and suggested that we can begin by adding the essential points inside than when just having a blank sheet of paper. Thus, the usage of sticker sheets containing layout and style elements (text boxes, title boxes, borders etc.) was suggested to help kickstart the making process.

• Sample templates with prompts

While the variety of designs excited the HNs, they felt there were too many directions to choose from.

HN2: "I like all these designs. I want to make something, but which design should I use?"

HN3: "Should I use all the designs here? Or can I use only two or three and make?"

HN1: "So, I can take this flower from the icons sheet, put it in this box and start writing? Is that it?"

A sample template was intentionally omitted to not restrict the usage of different elements together. However, after observing the

responses, a sample template was added to show how the elements could come together.

• Compiled Guide

All the materials mentioned above were compiled into a detailed how-to guide in three languages.

After sharing the layout elements and templates, the HNs, the field coordinator and the author discussed the possibility of uploading images of their posters to Papad as an additional layer to the annotations. The field coordinator also said she would like to promote friendly competition within the group to make more pamphlets. Even before the toolkit was finished, HN1 stated that she was inspired to make things independently and shared a pamphlet she tried to make (Figure 2 (c)).

4.1.2 Speculating futures for Papad. Previous discussions with the HNs had shown that they wanted their engagement with Papad to be more interesting and interactive. Some of the ideas shared by the HNs included one of them showing pictures they had taken and playing the recordings alongside them to suggest having voiceovers. Another HN talked about how multiple HNs have uploaded content about the same topic, and it would be helpful to stitch parts of those together. Based on these initial conversations, a new feature (simple audio-video editor) that uses Papad as a back-end framework was imagined through which the HNs could pick out annotations from across audios and stitch the fragments together to create new meaningful narratives. An activity was conducted with a group of 6-8 HNs to speculate various components that would form the new feature by building the interface and interactions together. The activity's main materials/ starting point was cut-outs from screenshots of the Papad interface so that the HNs could assemble them the way they wanted. Three simple steps were suggested as the framework of the activity.

- 1. Select Fragments-** This step involved asking the HNs to 'drag and drop' annotated fragment(s) from a recording into a 'jar' introduced to the interface. The HNs were prompted to 'fill up the jar' by searching the CHL for related recordings and fragments.
- 2. Edit, Stitch, Compile-** Here, the HNs were asked to put down their imaginations for how to manipulate the items within the jar to create some new content.
- 3. Share-** The final step was to upload the new content file to the CHL, download it locally, or use other channels to share it with the clients.



Figure 3: (a) HN added the first fragment to the jar (b) HN searched the CHL to fill the jar (c) HN's responses & ideas on content compilations

The HN's visions for compiling and creating new content based on the audio went beyond an audio editor that can help them combine things together. The HN's wanted various forms of compilations ranging from flyers to booklets to audio dramas. They talked about adding in their perspectives and missing information without taking away from the client's stories since they felt each of their experiences was valuable.

5 FINDINGS AND ANALYSIS

The key findings from these activities were interconnected and offered valuable insights regarding the interaction patterns that the HN's had with content and technology.

5.1 Annotations Framework

In the current framework for annotating, the HN's break the recording into fragments and restate what is being discussed. However, the larger goal of annotation is not to reiterate what is already being said but to highlight important areas by adding a layer of commentary to it. While the HN's naturally did this when they discussed among themselves, the form they had to fill to add an annotation to Papad had sections for time stamps, tags, long text descriptions and external files only. Typing long text is something that the HN's find difficult. They have tried using voice-to-text features but found the process more time-consuming. When it came to adding external files, they found it challenging to navigate file storage to access the correct file. As a result, the HN's kept their annotations brief even though it was clear that they wanted to add their own narratives and other missing information to the audio. Some others within COWHKE had arrived at similar conclusions and aimed to refine the annotations form. The proposed forms prompted the HN's to ask questions based on the audio and share related stories. They also tested a form where the entire audio is annotated without breaking into smaller time intervals. However, the new forms remained predominantly text dependent.

5.2 Organising and searching

During the speculative activity, when we were discussing searching through the CHL to build collections of related audios, the HN's talked about needing to know which information within the fragment could be used to build collections. We talked about using tags to see exact and related matches, and while this helped to move forward with the activity successfully, it is not the only method that

can be used to organise and search through content. The HN's knew that tags essentially captured what the recording was about. However, they often found it hard to break down the recordings into tags. Some of the tags were long and contained redundant words as well. For example, there were many tags like 'cancer bagge', 'BP bagge', etc. ('bagge' is the Kannada word for 'about'). Another difficulty with tagging came from spelling inconsistencies that arose due to using the English alphabet to spell Hindi and Kannada words. Finally, tags were also hard to visualise, making it hard to think through tags to make connections within the CHL.

5.3 Translating technology

Another key takeaway from the activities was that before creating a digital interface or tool to do something, it is good to test out the process by hand and then see what aspects of it could go on the software. While this sounds similar to paper prototyping, a popular part of the UX design process, traditionally, paper prototyping involves the representation of the screen and interface on paper to test the users' reaction to the same. There is a significant difference between using a paper representation of an interface and a paper representation of the process. In this case, in the speculative session, the informal goal was co-producing a paper prototype of a tool that would help the HN's compile content. However, since the process and methods of compilation were not finalised, the result of this activity was not a rough prototype co-created by the HN's that was expected but a collection of conceptual directions for the CHL ranging from ideas around different media for compiling content to discussing the need to have ways to add more layers to the recordings.

On the other hand, with the toolkit, what was co-designed was a paper representation of a process to create a single type of content compilation, i.e., a pamphlet. By using the toolkit to become familiarised with the process of creating/assembling graphical content using more minor elements, it may be easier to learn a digital tool that works on the same principles, such as Canva. Canva's enormous success stems from the fact that it provides us with layout and style elements as well as templates to help better organise content and make the world of visual design more accessible and less intimidating [2]. However, using Canva was still not the solution here because learning a new digital tool and process would be a significant cognitive load for the HN's.

6 RECOMMENDATIONS

Based on the findings and insights described earlier, future directions and frameworks for interactions within Papad and CHL were proposed as a short illustrated speculative fiction. The key features described in the story are-

1. **Annotations** - Moving away from text-heavy form, the new annotations framework suggests splitting the current descriptions field into three parts- 1) summary of what is said, 2) related stories, and 3) clarifying questions. Additional voice note features can be used alongside text to make it easier for HNs to share their thoughts. To ensure brevity, a time limit of one minute can be maintained for the description. Apart from having a voice note option, the similar experiences field has a set of suggested prompts from which HNs can select to respond to one. In case of insufficient information, the HNs choose from a list of possible questions. Response to the questions can be collectively discussed and added as a follow-up.
2. **Graph view of audio** - Graphs are a compelling way of visualising various connections. Several websites use graphs to help users navigate related content easily. One example is ConnectedPapers which lets the user see a graph of multiple academic papers related to the selected one [3]. For PAPAD, graphs can be utilised to visualise the various connections a given recording has to other files within the CHL. After adding annotation tags, the HNs are suggested other audios which they can choose to add as a link. Using these links and tags, a given recording can be visualised using a graph format.
3. **Information centres** - Owing to the lack of content and technology engagement within certain sections of the community, a possible avenue to explore would be the establishment of information desks at the local Public Health Centres (PHCs), Anganwadi centres, schools etc. Accessing CHL through the info desks would help the community reach information at places where they need it the most. The HNs could also display any pamphlets or flyers they created about different topics for people to take back with them.

The medium of an illustrated story was chosen so that the presented ideas could be easily and quickly read and grasped by the others in the COWHKI team. The story was first drafted in English, and work is being done to translate it into local languages to be shared with the HNs (this way, we can judge if their visions were accurately captured and to their satisfaction). One recommendation that was not covered in the story was about tags. As stated earlier, tags were hard to generate and visualise for the HNs and were prone to spelling mistakes. Thus, instead of requiring tags to be user-inputted, the platform can provide a list of suggested tags from which relevant ones can be chosen. Apart from this, tags need not be in single-word text form but can be accompanied by an illustration to make it easier to understand the tag's meaning.

7 CONCLUSION

The paper started by discussing the importance of local content in networks to empower users and make the Internet more relevant to them. We explore local content creation and dissemination in

the context of the COWHKI project in Channapatna. The methodology adopted was an incremental trust-building process involving the HNs in all phases of design. Through speculating future directions for Papad for enhanced content creation and dissemination, suggestions that bridge the gap between the HNs visions and the digital application were brought up. The co-designed toolkit aims to bridge the gap between physical and digital interfaces with a process that could eventually lead to the adoption of graphic design tools like Canva by the HNs. Both these interventions enhance CHL's content and make it more accessible to the community, promoting health awareness and literacy. However, infrastructure for healthcare needs to go beyond promoting health literacy and move towards helping the community make sense of the information as well [1]. While the HNs are involved in this aspect during their door services, looking into the engagement with the content post dissemination could be explored. The study approaches local content through the lens of one group of stakeholders, the HNs, who used their decade of experience as frontline workers to advocate methods for the community. The next step would be to study the larger Channapatna community's reaction to the same.

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